

Basic Comparison Test: Suppose $\sum a_i$ and $\sum b_i$ are series with positive terms, i.e. $a_n, b_n \geq 0$ for all n . Then

- If $\sum b_i$ converges and $a_n \leq b_n$ for all n , then $\sum a_i$ converges. $\rightarrow$ bigger series converges $\rightarrow$ smaller also does
- If $\sum b_i$ diverges and $a_n \geq b_n$ for all n , then $\sum a_i$ diverges. $\rightarrow$ smaller series diverges $\rightarrow$ bigger one also does

Example: Determine whether

$$\sum_{i=2}^{\infty} \frac{1}{\sqrt{i}-1}$$

$$a_n = \frac{1}{\sqrt{n}-1}$$

$$b_n = \frac{1}{\sqrt{n}}$$

converges or diverges.

If the smaller series diverges, which it does, then the larger series diverges

So $\sum_{i=2}^{\infty} a_n$ diverges

$$a_n \geq b_n$$

$\sum_{n=2}^{\infty} b_n$ (p-series w $p = \frac{1}{2}$)
diverges since $p < 1$

b_n 's denom is always bigger than a_n 's denom so b_n will be smaller

Example: Determine whether

$$\sum_{n=4}^{\infty} \frac{n^2-3}{n^5+4}$$

converges or diverges.

$$a_n = \frac{n^2-3}{n^5+4}$$

$$a_n \leq b_n$$

$$b_n = \frac{n^2}{n^5} = \frac{1}{n^3}$$

$$\sum_{n=4}^{\infty} \frac{1}{n^3}$$

p series where $p > 1$

$\hookrightarrow \sum_{n=4}^{\infty} b_n$ converges

Since $\sum_{n=4}^{\infty} b_n$ converges and $\sum_{n=4}^{\infty} a_n$ is smaller than $\sum_{n=4}^{\infty} b_n$ converges

Example: Determine whether

converges or diverges.

$$a_n = \sum_{n=3}^{\infty} \frac{9^n}{4^n - 1}$$

$$b_n \leq a_n$$

$$\sum_{n=3}^{\infty} \frac{9^n}{4^n - 1}$$

$$b_n = \sum_{n=3}^{\infty} \frac{9^n}{4^n}$$

$$b_n = \left(\frac{9}{4}\right)^n$$

$$r = \frac{9}{4} = 2.25$$

$$r \geq 1 \rightarrow \frac{9}{4} > 1$$

diverges as a geometric series

$\sum_{n=3}^{\infty} b_n$ is smaller and diverges so $\sum_{n=3}^{\infty} a_n$ diverges

Always write out concluding rationale

By comparison test, $\sum_{n=3}^{\infty} a_n$ diverges because $\sum_{n=3}^{\infty} b_n = \sum_{n=3}^{\infty} \left(\frac{9}{4}\right)^n$ diverges as a geometric series because $9/4 > 1$

Example: Determine whether

converges or diverges.

$$a_n = \frac{1}{n^n} \leq \frac{1}{2^n} = b_n$$

$$\sum_{n=2}^{\infty} b_n = \sum_{n=2}^{\infty} \left(\frac{1}{2}\right)^n \text{ converges as a geometric series because } \frac{1}{2} < 1$$

So by comparison test $\frac{1}{n^n}$ which is smaller also converges

$$n^n \geq n^2$$

$$\frac{1}{n^n} \leq \frac{1}{n^2}$$

$$\sum_{n=2}^{\infty} \frac{1}{n^2} \text{ converges as a p-series}$$

when doing comparison test you want to make a series that is a named series so you can easily determine

starting index only matters when determining what a series converges to

$n \geq 2$ counting from 2 to ∞

$n \geq 2^n$
n is at least 2 so n^n is at least 2^n

using 2^n does not help vs but we could use 2^n

When testing convergence or divergence we are looking @ big indices so starting 3
index does not matter for seeing if conv. or div. as long as you are not evaluating what it conv. to

Example: Determine whether

$$\sum_{k=3}^{\infty} \frac{(3k-1)(k^2-9)}{(k-3)(k^4+7)}$$

converges or diverges.

$a_n, b_n \geq 0$ for all n
If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$ where L is finite and nonzero, then $\sum a_n$ and $\sum b_n$ either both converge or both diverge

Limit Comparison Test: Suppose $\sum a_i$ and $\sum b_i$ are series with positive terms, i.e. $a_n, b_n \geq 0$ for all n . If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L \quad \leftarrow \text{Limit is a number that's not 0}$$

for some finite, nonzero L , then $\sum a_i$ and $\sum b_i$ either both converge or both diverge.

Example: Determine whether

$$\sum_{m=1}^{\infty} \frac{5}{\sqrt{m}+3}$$

converges or diverges.

Can't use basic
 b_n is bigger
+ since b_n diverges that tells us nothing

$$b_m = \frac{5}{\sqrt{m}} \rightarrow \sum_{m=1}^{\infty} b_m \text{ diverges as a } p\text{-series because } p < 1$$

$$\lim_{m \rightarrow \infty} \frac{a_m}{b_m} = \lim_{m \rightarrow \infty} \left(\frac{5}{\sqrt{m}+3} \right) \left(\frac{\sqrt{m}}{5} \right) = \lim_{m \rightarrow \infty} \frac{\sqrt{m}}{\sqrt{m}+3} = 1$$

By limit comparison test,

$$\sum_{m=1}^{\infty} a_m \text{ diverges}$$

So determine behavior of b_n and then based on limit we choose if they have the same behavior
If limit is 0 or ∞ , change what b_n is because result is inconclusive. 0 or use a different test

You can also
do $\lim_{m \rightarrow \infty} \frac{\sqrt{m}}{\sqrt{m}+3} = 5$
 $\left(\lim_{m \rightarrow \infty} \left(\frac{5}{\sqrt{m}+3} \right) \left(\frac{\sqrt{m}}{2} \right) \right)$

Why is this true

Example: Determine whether

$$\sum_{n=2}^{\infty} \frac{n^4 + 3n^2}{n^7 + 17n^3 + 2n + 8}$$

converges or diverges.

$b_n = \frac{1}{n^3}$ $\sum_{n=2}^{\infty} b_n$ converges as a p-Series ($p = 3 > 1$)

$$\frac{a_n}{b_n} = \left(\frac{n^4 + 3n^2}{n^7 + 17n^3 + 2n + 8} \right) \cdot \left(\frac{n^3}{1} \right) = a_n \div \frac{1}{n^3} = a_n \times \frac{n^3}{1}$$

$$= \frac{n^7 + 3n^5}{n^7 + 17n^3 + 2n + 8}$$

Highest degrees are same so
lim = 1?

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1$$

By limit comparison test, $\sum a_n$ converges

Example: Determine whether

$$\sum_{k=1}^{\infty} \frac{1}{k^{1+1/k}}$$

converges or diverges.

$$a_k = \frac{1}{k^{1+1/k}}$$

$$b_k = \frac{1}{k}$$

$\sum_{k=2}^{\infty} b_k$ diverges as a harmonic series diverges

So if limit $\neq 0$ and limit of ratio $\neq \infty$, a_k also diverges

$$\lim_{k \rightarrow \infty} \frac{a_k}{b_k}$$

$$= \lim_{k \rightarrow \infty} \frac{1}{k^{1+1/k}} \cdot \left(\frac{k}{1} \right) = \lim_{k \rightarrow \infty} \frac{k}{k \cdot k^{1/k}} = \lim_{k \rightarrow \infty} \frac{1}{k^{1/k}}$$

By function methods, $\lim_{x \rightarrow \infty} \frac{1}{x^{1/x}} = \lim_{x \rightarrow \infty} \frac{1}{e^{\ln(x)/x}} = \lim_{x \rightarrow \infty} \frac{1}{e^{\frac{\ln(x)}{x}}} = \frac{1}{e^{\lim_{x \rightarrow \infty} \frac{\ln(x)}{x}}} = \frac{1}{e^0} = 1$

$L \neq 0$, $L \neq \infty$ by limit comparison test, $\sum_{k=2}^{\infty} a_k$ diverges